Ashik Ahamed

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RESEARCH INTERESTS

Machine learning and explainable AI for climate extremes and heat-health risk, with a focus on combining ground-station time series with remote sensing and spatial data for spatiotemporal prediction.
Keywords: Remote Sensing and Geospatial Analysis · Spatiotemporal Machine Learning · Time-Series Forecasting · Explainable AI (SHAP, LIME) · Climate Extremes · Heat-Health Risk · Air Quality Prediction

EDUCATION

American International University–Bangladesh (AIUB) 2021 – 2025
B.Sc. in Computer Science and Engineering (Major: Information Systems), Dhaka, Bangladesh

- CGPA: 3.89 / 4.00; Dean's List (4 times)
- Coursework: Machine Learning, Data Science, Probability & Statistics, Artificial Intelligence & Expert Systems
- English: IELTS Academic, Overall Band 7.0 (2026)

PUBLICATIONS

A. Ahamed (first author), et al. *Enhancing Heatwave Prediction Using Explainable AI: Feature Selection and Model Optimization for Improved Forecasting.* International Conference on Frontiers in Computing and Systems (COMSYS 2025). Accepted, to appear in proceedings (Springer Nature); presented at the conference.

- Built a 10-year (2010–2020) daily climate dataset for Dhaka; used SHAP-based feature selection to identify key predictors of daily maximum temperature
- Benchmarked GRU, LSTM, RNN and XGBoost using MSE, MAE and R^2

RESEARCH EXPERIENCE

Research Assistant May 2025 – Mar 2026
Supervisor: Nuzhat Tabassum, Assistant Professor, Dept. of Computer Science

- Cleaned, preprocessed and engineered features from raw multi-year climate data for heatwave and temperature prediction research
- Implemented and evaluated machine learning and time-series models (tree ensembles, GRU, LSTM) with explainability analysis
- Prepared technical documents in LaTeX and presentations to communicate research findings

Research Student, Elite Research Lab Sep 2026 – Present

- Structured mentorship in manuscript writing and framing research contributions under professor and PhD-level mentors
- Work in mentor-led research groups on one paper at a time across domains (mostly environmental topics so far); currently co-authoring a team paper targeting *IEEE Transactions on Intelligent Transportation Systems*, as part of a group paper-writing competition

MANUSCRIPTS IN PREPARATION

Time-Series Analysis of Climate Extremes: A Multi-City Heatwave Prediction and Explainability Study. With N. Tabassum (AIUB).

- Three-phase pipeline (feature-based prediction, time-series forecasting, explainability) on a 14-year (2010–2024) daily temperature dataset for Dhaka
- Nine feature-selection-driven ML models benchmarked; champion Stacking model reached MAE 0.66°C, R^2 0.959. Classical and deep time-series models benchmarked (ARIMA, SARIMA, LSTM, BiLSTM, RNN, GRU, CNN-LSTM, PyCaret); champion GRU reached R^2 0.78
- LIME (local) and Morris Sensitivity Analysis (global) identified dry-bulb temperature as the dominant driver
- Extended to Rajshahi, Chittagong and Sylhet: strong regression generalization (R^2 0.68–0.98), with city-dependent classification performance

RESEARCH PROJECTS

ML Framework for Next-Day Heat Risk Prediction in Bangladesh Jun 2026 – Present
Independent research project

- Predict next-day heat risk level (Safe, Caution, Danger, Extreme Danger) for eight Bangladesh cities from previous-day meteorological features (MinTemp, Cloud, Rainfall, DryBulbTemp, Sunshine, WindSpeed), using 11 years of weather data
- Risk labels derived from the Heat Index (Rothfusz equation) using MaxTemp and Humidity, both excluded from model inputs to prevent data leakage
- Benchmarking XGBoost, Random Forest and LSTM for multi-class prediction; SHAP to identify city-specific risk factors; proposing an early-warning framework

Explainable AI in Education for Personalized Learning 2024 – 2025
Python, scikit-learn, XGBoost, SHAP

- XAI framework to predict student GPA and classify stress levels; SHAP identified study hours, sleep and physical activity as key predictors (SVC: 95% accuracy, F1 0.94; Linear Regression R^2 0.538)

Other Technical Projects

- **Face Recognition System** (Python, TensorFlow): CNN-based recognition with augmentation, evaluated by accuracy, precision, recall and confusion matrix
- **Data Visualization in Pharmaceutical Evaluation** (R, ggplot2): exploratory analysis and publication-quality visualizations of drug-evaluation data

TECHNICAL SKILLS

Programming: Python, R, SQL **Tools:** Power BI, MySQL, Jupyter, LaTeX
ML/DL: scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch
Time series: ARIMA, SARIMA, LSTM, GRU, BiLSTM, CNN-LSTM, PyCaret
Explainability: SHAP, LIME, Morris Sensitivity Analysis

OTHER PROFESSIONAL EXPERIENCE

Content Developer, General R&D Dept., Lecture Publications, Dhaka Apr 2026 – Present

- Develop academic book content, restructuring complex material into clear explanations aligned with curriculum standards

Data Analyst, DC IT Management, Dhaka Sep 2025 – Feb 2026

- Automated SQL–Python reporting pipelines (monthly reporting time reduced by 30%); built Power BI dashboards used by sales managers

Junior Data Analyst, Super Star Group, Dhaka Dec 2024 – Apr 2025

- Applied ML models to business data to identify trends and support decision-making

REFERENCES

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